

```

In [1]: #step1: load dataset
import pandas as pd
df = pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
print(df.head())

#2 Renaming Columns
#Check current Column Names
print(df.columns)

df.rename(columns={
    'Student ID':'Student_ID',
    'Student Name ':'Student_Name',
    ' Department':'Department'
},inplace=True)
print(df.columns)

#3.Changing Data Types
#Check Data Type
print(df.dtypes)

#Convert Student_ID to String
df['Student_ID'] = df['Student_ID'].astype(str)

#Convert Age to Integer
#First handle Missing values
df['Age'] = df['Age'].fillna(0)
df['Age'] = df['Age'].astype(int)

```

	Student ID	Student Name	Department	Age	Salary	Email
0	101	student1	IT	20.0	23075.0	student1@gmail.com
1	102	student2	ECE	18.0	35929.0	student2@gmail.com
2	103	STUDENT3	EEE	22.0	29718.0	student3@gmail.com
3	104	student4	MECH	20.0	21009.0	student4@gmail.com
4	105	Student5	CSE	19.0	47113.0	student5@gmail.com

```

Index(['Student ID', 'Student Name ', ' Department', 'Age', 'Salary', 'Email'], dtype=
object)
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype
=object)
Student_ID      int64
Student_Name    object
Department      object
Age             float64
Salary          float64
Email           object
dtype: object

```

```

In [2]: #Check
print(df.dtypes)

#4.Detecting Dulicate Records
#Find Duplicates
duplicates = df.duplicated()
print(duplicates)

```

```
print(df.duplicated().sum())
```

Student_ID	object
Student_Name	object
Department	object
Age	int64
Salary	float64
Email	object

dtype: object

0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False
13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False
23	False
24	False
25	False
26	False
27	False
28	False
29	False
30	False
31	False
32	False
33	False
34	False
35	False
36	False
37	False
38	False
39	False
40	False
41	False
42	False
43	False
44	False
45	False
46	False
47	False
48	False

```
49    False
50    False
51    False
dtype: bool
0
```

```
In [3]: #4.Detecting Duplicate Records
        #Find Duplicates
        duplicates = df.duplicated()
        print(duplicates)

        #Count Duplicates
        print(df.duplicated().sum())

        #5.Drop Duplicates Records
        #Before
        print(df.shape)

        #Remove Duplicates
        df.drop_duplicates(inplace=True)

        #After
        print(df.shape)
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
10   False
11   False
12   False
13   False
14   False
15   False
16   False
17   False
18   False
19   False
20   False
21   False
22   False
23   False
24   False
25   False
26   False
27   False
28   False
29   False
30   False
31   False
32   False
33   False
34   False
35   False
36   False
37   False
38   False
39   False
40   False
41   False
42   False
43   False
44   False
45   False
46   False
47   False
48   False
49   False
50   False
51   False
dtype: bool
0
(52, 6)
(52, 6)
```

```
In [4]: #4.Detecting Duplicate Records
#Find Duplicates
duplicates = df.duplicated()
print(duplicates)

#Count Duplicates
print(df.duplicated().sum())

#5.Drop Duplicate Records
#Before
print(df.shape)

#Remove Duplicates
df.drop_duplicates(inplace=True)

#After
print(df.shape)
#6.String Operations
#string methods work only on text columns.
#Remove extra spaces
df['Student_Name'] = df['Student_Name'].str.strip()
#convert to uppercase
df['Student_Name'] = df['Student_Name'].str.upper()
print(df['Student_Name'].head())
#convert to TitleCase
df['Student_Name'] = df['Student_Name'].str.title()
print(df['Student_Name'].head())
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
10   False
11   False
12   False
13   False
14   False
15   False
16   False
17   False
18   False
19   False
20   False
21   False
22   False
23   False
24   False
25   False
26   False
27   False
28   False
29   False
30   False
31   False
32   False
33   False
34   False
35   False
36   False
37   False
38   False
39   False
40   False
41   False
42   False
43   False
44   False
45   False
46   False
47   False
48   False
49   False
50   False
51   False
dtype: bool
0
(52, 6)
(52, 6)
```

```

0    STUDENT1
1    STUDENT2
2    STUDENT3
3    STUDENT4
4    STUDENT5
Name: Student_Name, dtype: object
0    Student1
1    Student2
2    Student3
3    Student4
4    Student5
Name: Student_Name, dtype: object

```

```

In [5]: #Replace Text
        #Replace department name.
        df['Department']=df['Department'].str.replace(
            'CSE',
            'Computer Science'
        )
        print(df['Department'].head())

        #Replace Multiple Values - No Need str
        df['Department']=df['Department'].replace({
            'CSE':'Computer Science',
            'ECE':'Electronics',
            'IT':'Information Technology'
        })
        print(df['Department'].head())

```

```

0          IT
1          ECE
2          EEE
3          MECH
4  Computer Science
Name: Department, dtype: object
0  Information Technology
1          Electronics
2          EEE
3          MECH
4          Computer Science
Name: Department, dtype: object

```

```

In [6]: #Replace Text
        #Replace department name.
        df['Department']=df['Department'].str.replace(
            'CSE',
            'Computer Science'
        )
        print(df['Department'].head())

        #Replace Multiple Values - No Need str
        df['Department']=df['Department'].replace({
            'CSE':'Computer Science',
            'ECE':'Electronics',
            'IT':'Information Technology'
        })

```



```

})
print(df['Department'].head())

```

```

0    Information Technology
1           Electronics
2                EEE
3                MECH
4       Computer Science
Name: Department, dtype: object
0    Information Technology
1           Electronics
2                EEE
3                MECH
4       Computer Science
Name: Department, dtype: object

```

```

In [12]: #String Length
df['Name_length']=df['Student_Name'].str.len()
print(df[['Student_Name','Name_length']].head())
#Extract First Character
df['First_letter']=df['Student_Name'].str[0]
print(df[['Student_Name','First_letter']].head())

```

```

Student_Name  Name_length
0    Student1           8
1    Student2           8
2    Student3           8
3    Student4           8
4    Student5           8
Student_Name  First_letter
0    Student1            S
1    Student2            S
2    Student3            S
3    Student4            S
4    Student5            S

```

```

In [13]: #Replace Text
#Replace department name.
df['Department']=df['Department'].str.replace(
    'CSE',
    'Computer Science'
)
print(df['Department'].head())

```

```

Student_Name  Name_length
0    Student1           8
1    Student2           8
2    Student3           8
3    Student4           8
4    Student5           8
Student_Name  First_letter
0    Student1            S
1    Student2            S
2    Student3            S
3    Student4            S
4    Student5            S

```

